

ESE DEC 2020_TY COMP (Elective) Soft Computing_21-12-2020

* Required

1. Email address *

2. Enter your full name *

3. Enter your roll number *

Questions

4. Core of soft Computing is

1 point

Mark only one oval.

- ☐ Fuzzy Computing, Neural Computing, Genetic Algorithm
- ☐ Fuzzy Networks and Artificial Intelligence
- ☐ Artificial Intelligence and Neural Science
- ☐ Neural Science and Genetic Science

5. What is perceptron?

1 point

Mark only one oval.

- ☐ a single layer feed-forward neural network with pre-processing
- ☐ an auto-associative neural network
- ☐ a double layer auto-associative neural network
- ☐ a neural network that contains feedback

6. What is an auto-associative network?

1 point

Mark only one oval.

- ☐ a neural network that contains no loops
- ☐ a neural network that contains feedback
- ☐ a neural network that has only one loop
- ☐ a single layer feed-forward neural network with pre-processing

7. A 4-input neuron has weights 1, 2, 3 and 4. The transfer function is linear with the constant of proportionality being equal to 2. The inputs are 4, 10, 5 and 20 respectively. What will be the output?

1 point

Mark only one oval.

- ☐ 238
- ☐ 76
- ☐ 119
- ☐ 123

8. ._____deals with uncertainty problems with its own merits and demerits 1 point

Mark only one oval.

- ☐ neuro –fuzzy
- ☐ neuro-genetic
- ☐ fuzzy –genetic
- ☐ done

9. _____function is a continuous function that varies gradually between the asymptotic values 0 and 1 or -1 and +1 1 point

Mark only one oval.

- ☐ Activation function
- ☐ Thresholding function
- ☐ Signum function
- ☐ Sigmoidal function

10. ADALINE stands for _____ 1 point

Mark only one oval.

- ☐ Adaptive Linear Neural Element Network
- ☐ Adaptive Line Neural Network
- ☐ Adapt Line Neural Element Network
- ☐ Adaptive Linear Neural Network

11. Linear Activation Operator equation is _____

1 point

Mark only one oval.

- ☐ $O=gl, g=\tan\phi$
- ☐ $O=gl, g=\sin\phi$
- ☐ $O=gl, g=\cos\phi$
- ☐ $O=gl, g=-\tan\phi$

12. Reinforced learning also known as _____

1 point

Mark only one oval.

- ☐ Output based learning
- ☐ Back propagation learning
- ☐ Error based learning
- ☐ None

13. _____ learning follows the “Winner takes all” strategy

1 point

Mark only one oval.

- ☐ Stochastic learning
- ☐ Competitive learning
- ☐ Hebbian learning
- ☐ BackPropagation learning

14. If the associated pattern pairs (x,y) are different and if the model recalls a y given an x or vice versa, then it is termed as _____

1 point

Mark only one oval.

- ☐ Auto associative memory
- ☐ neuro associative memory
- ☐ Hetero associative memory
- ☐ None

15. _____ uses bipolar coding

1 point

Mark only one oval.

- ☐ Fabric defect identification
- ☐ Design of Journal Bearing
- ☐ Recognition of Characters
- ☐ Classification of soil

16. BAM was introduced by _____

1 point

Mark only one oval.

- ☐ Cruz
- ☐ Stubberd
- ☐ Kosko
- ☐ Rosenbatt

17. _____ of the network means that a pattern should not oscillate among different cluster units at different stages of training 1 point

Mark only one oval.

- ☐ Stability
- ☐ Mobility
- ☐ Versatility
- ☐ Placticity

18. In _____ learning the weights are adjusted only when the external input matches one of the stored prototypes 1 point

Mark only one oval.

- ☐ Supervised
- ☐ UnSupervised
- ☐ Match-based
- ☐ None

19. _____ obtains centre of area occupied by the fuzzy set 1 point

Mark only one oval.

- ☐ center
- ☐ center of gravity
- ☐ center of area
- ☐ center point

20. The _____ of an organism is measured by means of success of organism in life 1 point

Mark only one oval.

- ☐ Strength
- ☐ fitness
- ☐ Gene
- ☐ Chromosome

21. _____ is a way of representing individual genes

1 point

Mark only one oval.

- ☐ conversion
- ☐ encoding
- ☐ coding
- ☐ decoding

22. Knowledge-based evaluation and earthquake damage evaluation is application of _____

1 point

Mark only one oval.

- ☐ fuzzy-backpropagation
- ☐ neuro-fuzzy
- ☐ fuzzy
- ☐ none

23. In the neuron, attached to the soma are long irregularly shaped filaments called _____

1 point

Mark only one oval.

- ☐ Dendrites
- ☐ Axon
- ☐ Synapse
- ☐ Cerebellum